Why we stopped optimizing for competition results

The case for treating tournaments as a side effect of a healthy program, not the goal of one.

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For our first three years, the calendar at Growing Up with Robotics was organized around competitions. Not loosely organized. Built around. Every cohort had a tournament on the horizon, every curriculum sequence ended in a qualifier, every Saturday afternoon was implicitly being graded against how ready the team would be when the judges arrived. We thought this was good for the kids. We thought a deadline produced focus, and a trophy produced motivation, and a ranking produced a way to know whether we were getting better as a program. We thought all of these things for longer than we should have.

I want to be careful here, because I am not going to argue that competitions are bad. They are not. A well-run tournament is one of the most concentrated learning experiences a young engineer will ever get, and watching a thirteen-year-old explain her code to a judge for the first time is genuinely one of the things that makes this work worth doing. The argument is narrower than that. The argument is that optimizing for competition results, as the primary metric of a youth robotics program, quietly destroys most of what makes those programs worth running in the first place.

What the trophy actually selects for

The first thing you notice, if you watch a program over enough seasons, is what competition optimization actually rewards. It does not reward the students who learn the most. It rewards the students who learned the most before they got to you. The eighth grader whose father is an electrical engineer is going to outperform the eighth grader whose father drives a delivery truck, not because she is smarter, not because she is working harder, but because she walked in the door with three years of garage-tinkering background that the other student does not have. If your program is being graded by tournament outcomes, your program will, slowly and without anyone deciding to do this, start to concentrate its resources on the first student and not the second.

This happens in small ways. The strongest builder gets to drive at competitions, so the strongest builder gets the most practice driving, so the strongest builder becomes the strongest driver. The student who picks up code fastest gets pulled into the autonomous routines, so that student writes more code, so that student keeps being the one who writes the code. By the third month of the season, your team has stratified itself into a small group of high-performers who are doing the interesting work and a larger group of support roles who are essentially watching. None of this is anyone's fault. It is what the competition incentive does to a team if you let it run unchecked.

The students who came in with the least are the ones who get the least out. This is the exact opposite of what an outreach program is supposed to do.

The metrics that compete with each other

The second thing you notice is that competition performance and learning are correlated but not the same thing, and when they conflict, the program almost always picks performance. A student who has been struggling with a sensor for three weeks is finally close to figuring it out. There is a competition in eleven days. The mentor's choice is to let her keep struggling, in which case she might not have it working by Saturday, or to step in and fix it for her, in which case the team has a working sensor and she has learned slightly less than she would have if she had finished the struggle. If your program is graded on Saturday's performance, you fix the sensor. If your program is graded on her learning, you don't. Most programs fix the sensor, and tell themselves they will let her struggle next time, and next time there is another competition in eleven days, and so on.

We were doing this. I do not want to pretend we were not. We have notebooks full of sessions where, looking back at them now, you can see the moment where a mentor took the screwdriver out of a student's hand because there was no time, and that moment was the actual lesson the student learned that day. The lesson was that when it matters, an adult will do it for you. That is not a lesson we wanted to teach. We taught it anyway, because the tournament was on the calendar and the tournament does not care about your pedagogy.

There is also the matter of which students get put forward at all. Competition rules cap how many students from a team can compete, or drive, or present to judges. The students who get those slots are the ones most likely to win. The students who don't get those slots are the ones who would have benefited most from the experience. The program ends up giving the most growth opportunity to the students who needed it least, and the least to the ones who needed it most. Again, no one decides to do this. It is what the structure does.

The retention problem nobody talks about

The third thing, and this is the one that finally changed how we thought about the whole question, is what happens to students after the competition season ends. We started tracking, around year two, what fraction of our students stayed engaged with robotics through the off-season. The number was bad. Not catastrophic, but bad enough that we had to look at it.

What we found was that the students who had been on the competing core were largely still engaged. They had identities now. They were robotics kids. They were going to keep being robotics kids. The students who had been on the broader team, the ones who had built and tested but not competed, mostly drifted away. They had spent six months helping a team they were not really part of win something they did not really get credit for, and then the season ended and there was nothing left for them. We had treated them as supporting infrastructure for the trophy. They had figured this out before we did, and they left.

This is the part I want people running programs to take seriously, because it is the part the competition-focused model cannot fix from inside itself. You can run the best, most equitable, most carefully managed tournament-prep program in the world, and you will still have this problem, because the structure of competition produces a small group of people who are doing the thing and a larger group of people who are watching it happen. The watchers will leave. They are right to leave. There is nothing for them in the model we built.

What we changed, and what it cost us

The change we made, starting in our fourth year, was to stop building cohorts around competitions and start building them around what we now call demonstrable artifacts. Every student, by the end of a twelve-week cohort, has to produce something they can show their family, explain to a stranger, and modify in front of a mentor without instructions. The artifact does not have to be impressive. It has to be theirs. A line follower that works on one specific track they designed. A sorting robot that handles three colors of blocks. A remote-controlled car with a custom button layout they argued for. The criterion is not how it compares to other students' artifacts. The criterion is whether the student can stand next to it and own it.

Competitions still exist in our calendar. They are now opt-in, organized as a separate track, and explicitly framed as one of several ways to extend what a student has learned. About a third of our students opt in, in any given season. The other two-thirds work on personal projects, on outreach to younger cohorts, on extending their artifacts in directions the competition format wouldn't have allowed. Our trophy count is lower than it used to be. Our retention through the off-season has roughly doubled. The students who do compete are, by every measure we can track, more prepared than they were under the old model, because they got there through their own pull rather than the program's push.

The cost was real. Our most competitive families left in the first year of this change. They were not wrong to leave. They had signed up for a program that produced trophies, and we had become a program that produced something else. Some of the mentors left too. Coaching to a competition is a more legible job than coaching to a portfolio, and a few of our strongest mentors found the new model frustrating in ways the old one had not been. I do not blame them either. We told them the rules of the work had changed, and for some people, the new rules were not the work they had signed up to do.

What the trophy was actually for

If I had to say the thing I most underestimated when we started, it would be how much the trophy was doing for us, as program runners, that had nothing to do with the students. The trophy was a way of telling sponsors that the program worked. The trophy was a way of telling parents that their child was in good hands. The trophy was a way of telling ourselves, on the hard weeks, that we were not wasting our time. None of those reasons are about whether a ten-year-old understood what a motor encoder does. All of them are about us.

The honest version of why most programs optimize for competition results is that the results give the adults something to point at. A graph of retention numbers and unprompted demos does not photograph as well as a team holding a banner. It does not show up in a sponsor deck the same way. It does not fit on a parent's refrigerator. The competition trophy is, more than anything, a credibility shortcut for the people who need to explain the program to people who are not in the room when the actual learning happens.

We are still working out how to replace that. Some of it is just doing the harder work of explaining the program well, in writing, in person, in the kind of conversations where you have to sit with someone for twenty minutes and walk them through what a portfolio means and why it is harder to fake than a ranking. Some of it is having older students tell their own stories, which is more persuasive than anything we could write. Some of it is accepting that we will be a less impressive-sounding program for a while, and that this is the price of being a more honest one.

If you are running a program that has organized itself around competitions, I am not going to tell you to stop. The format works for some kinds of students and some kinds of mentors, and the energy a tournament generates is real and valuable. What I would ask you to do is look at the students who are not in your competing core. Look at the ones whose hands are not on the robot at the qualifier. Ask what your program is for them, specifically, separate from what they contribute to the team's results. If the answer is that the program is mostly for someone else, and they are mostly there to help that someone else succeed, that is the thing to fix. The trophy is not going to fix it for you. We tried for three years.

Growing Up with Robotics runs cohort-based robotics programs for students aged 8 to 16. To learn more about our portfolio model or to sponsor a cohort, visit our website.